For the April 13 submission

Our MVP is a basic web app called PantryPal where users enter a list of ingredients they already have, and the system processes that input to generate about three relevant meal suggestions. Each suggestion includes a short description of the meal and highlights any additional ingredients that might be needed to complete it. It focuses on simple ingredient matching using a recipe dataset, without advanced features, just enough functionality to help users quickly decide what they can cook with what they already have.

The main technical risks and bottlenecks are mostly around data quality, recommendation quality, and keeping the system simple to use. The project notes that recipe datasets may need a lot of cleaning before they can be used well, which could slow development. There is also a risk that the meal suggestions the system generates may not always be realistic, useful, or appealing, especially when users enter messy ingredient lists. Another bottleneck is making sure the app can correctly interpret casual, flexible user input and still match it to recipes in a reliable way. Since the MVP depends on ingredient matching and basic suggestions, the team will need to balance accuracy with simplicity so users can get fast results without having to enter too much information.

What works already:

- Our code runs and produced a good output as well as accepts a lot of different inputs.

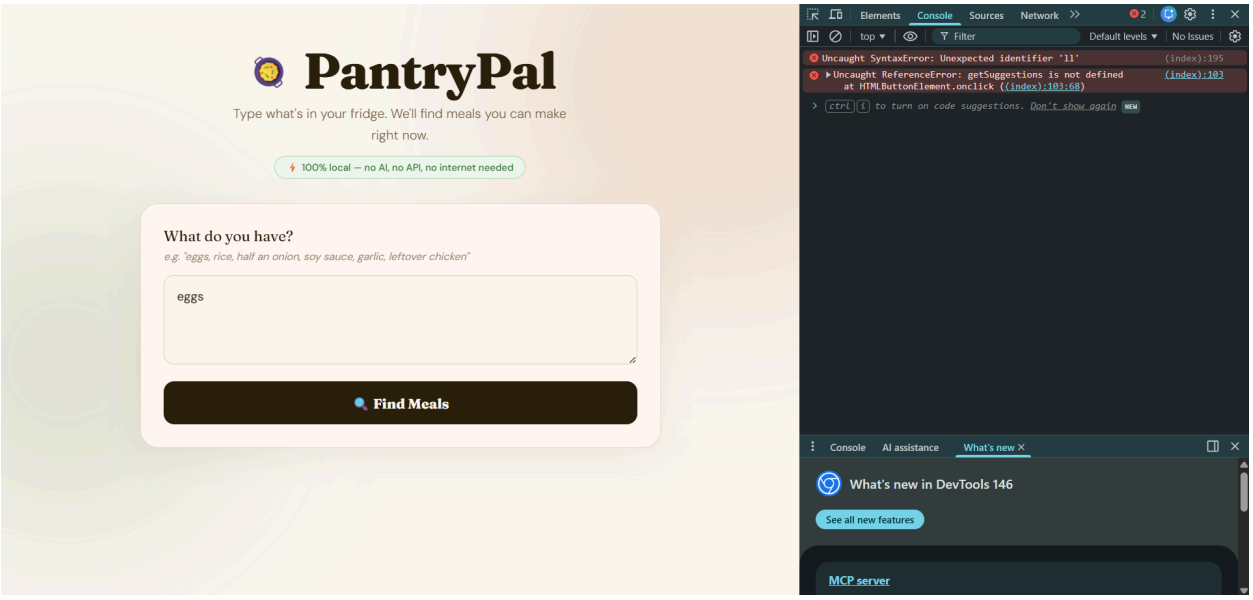
What doesn't work yet:

- We do not have the AI itself running, it is just coded to run output based on what we tell it to.

What will be completed:

- Updating our code to run and startup the AI to produce outputs on its own.

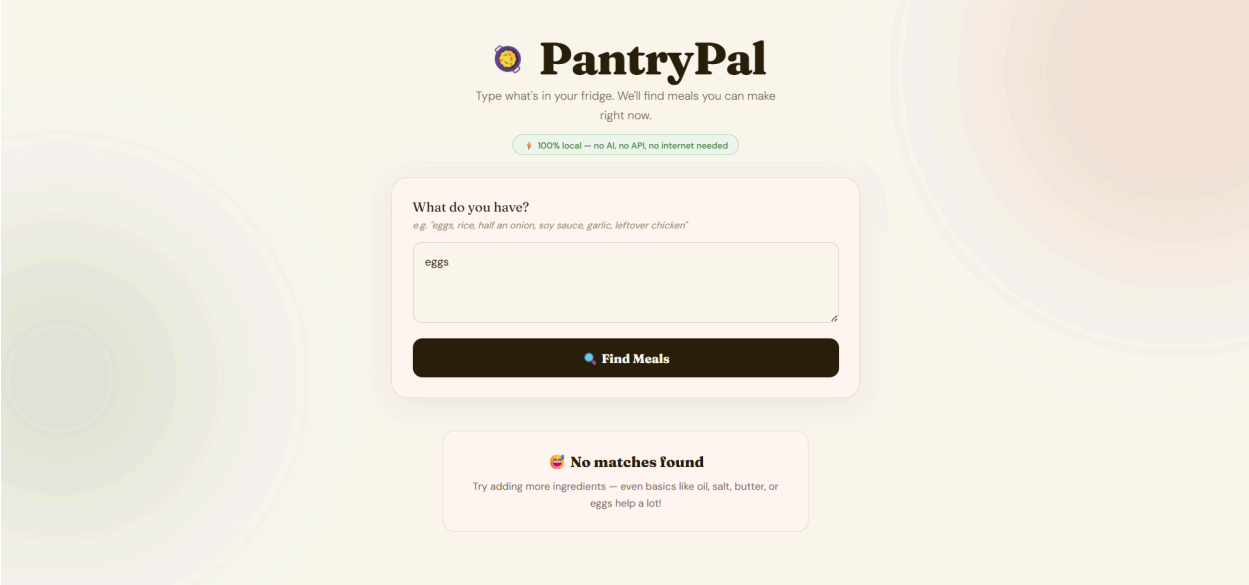
Failure Case:

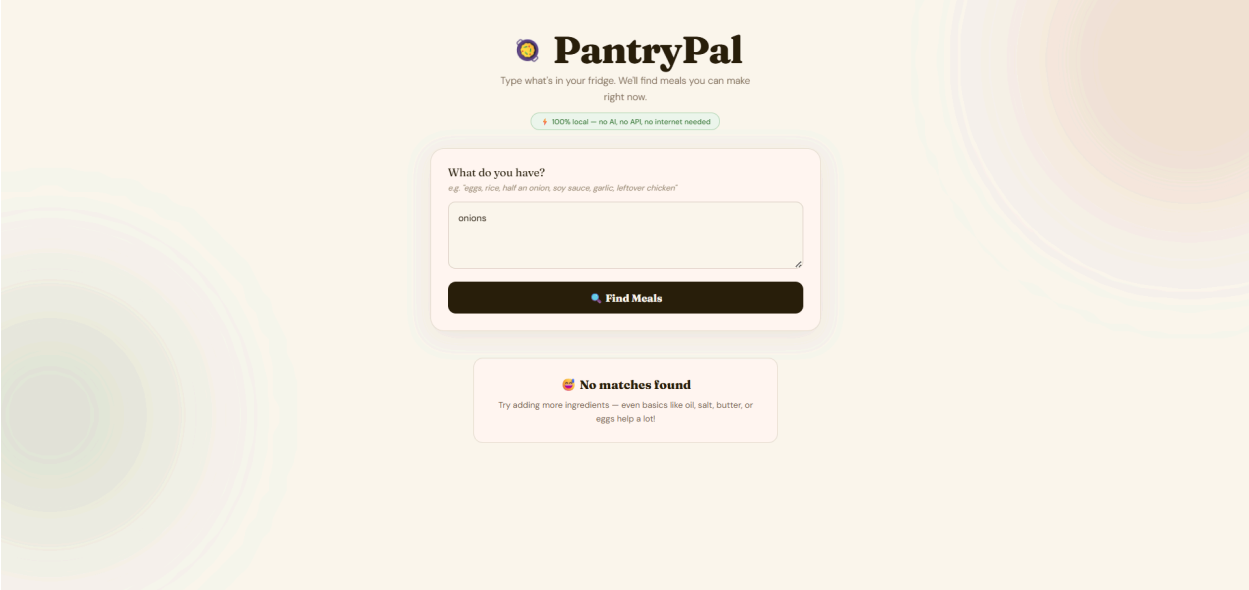


Description: There was an error in the code involving an apostrophe in the word “You’ll” which caused the code not to run properly. When the word “eggs” was typed into the user input, no recipes were returned.

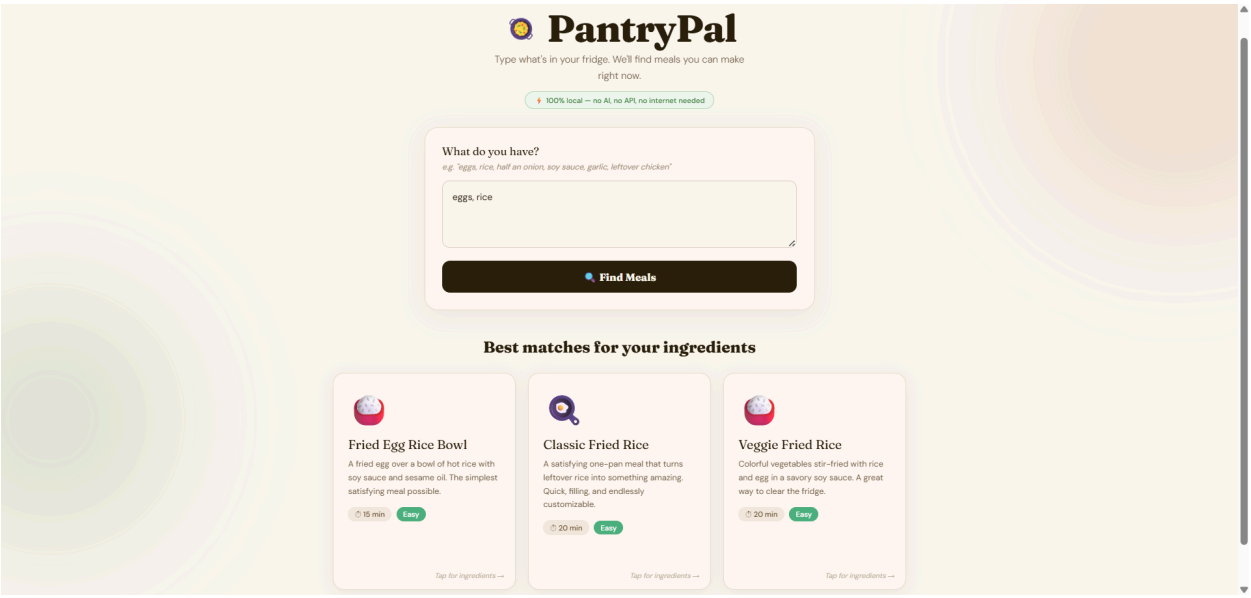
First iteration: Used Claude to create a PantryPal website using no AI, just hardcoded an ingredient list into the code as well as a recipe list that corresponded to ingredients.

Failure Case:





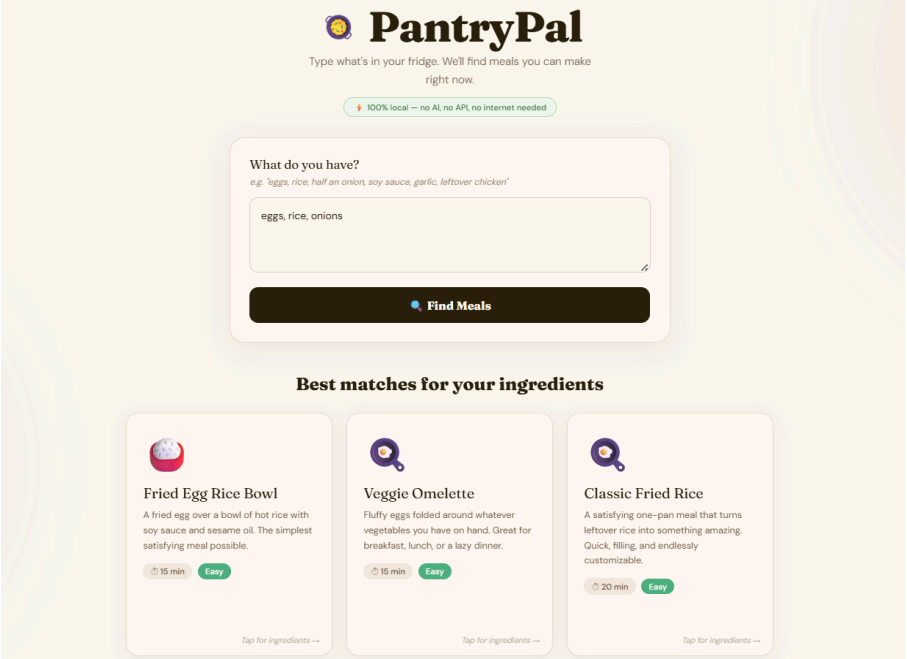
Description: The app fails when only the word “eggs” is inputted into the user input. The app also fails when only the word “onions” are inputted. This is because the code requires more than one ingredient in order for it to find matches. Next, inputting two ingredients will be tried.



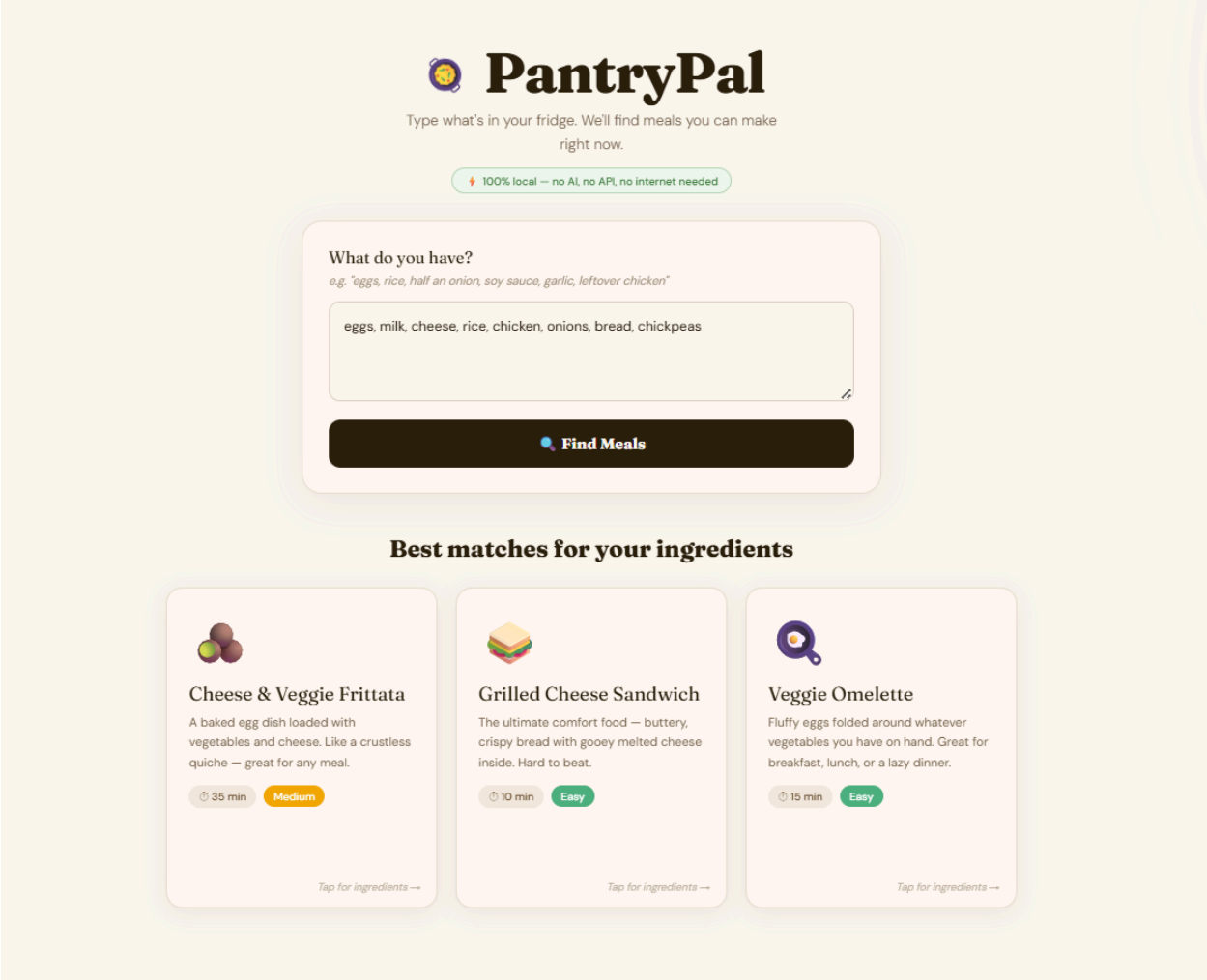
Two ingredients seemed to work well, we were given three recipes.

The required ingredients seem to be a lot for all of them. The minimum amount of required ingredients that are not present across all of these recipes is 4, which may be an issue to the user.

Three ingredients were tested which works well:



Eight ingredients were tested:



When there are ingredients that are not hardcoded, it seems that the code ignores them and just focuses on the ingredients that are hardcoded and matches them with recipes that are available in the database.